

MLOps Assignment 2 Final Report

Hugging Face Fine-Tuning, Experiment Tracking & Model Deployment

Student Name: Isha Rathod

Program: PGD AI Program | IIT Jodhpur

Assignment: MLOps Assignment 2

1. Introduction

The objective of this assignment was to implement a complete MLOps workflow using Kaggle GPU infrastructure, Hugging Face Transformers, Weights & Biases (W&B;), and Hugging Face Hub. The workflow included dataset preparation, model fine-tuning, experiment tracking, evaluation, and deployment.

2. Model Selection Rationale

DistilBERT (distilbert-base-cased) was selected because it is lightweight, faster, and requires less GPU memory while maintaining strong language understanding capabilities. It is ideal for Kaggle's free GPU environment and integrates smoothly with the Hugging Face ecosystem.

3. Kaggle Training Setup

The project was implemented using Kaggle Notebooks with GPU acceleration enabled using T4 GPU. Internet access was enabled for downloading packages and pushing models to Hugging Face Hub. Kaggle Secrets were used to securely store WANDB_API_KEY and HF_TOKEN.

4. Workflow Implementation

The workflow included importing the dataset, tokenizing text data, loading the DistilBERT tokenizer and model, training using the Hugging Face Trainer API, tracking experiments with W&B;, evaluating the model, and deploying the trained model to Hugging Face Hub.

5. Workflow Steps

- Configured Kaggle Notebook environment.
- Enabled GPU and Internet access.
- Configured Kaggle Secrets securely.
- Loaded and tokenized AG News dataset.
- Loaded DistilBERT tokenizer and model.
- Configured Hugging Face TrainingArguments.
- Tracked training metrics using W&B;.
- Evaluated the trained model.

- Uploaded model and tokenizer to Hugging Face Hub.

6. Training & Evaluation Results

The model was trained successfully for 3 epochs using Kaggle GPU resources. Weights & Biases automatically tracked metrics, hyperparameters, and training logs.

Metric	Score
Accuracy	0.86
F1 Score	0.86
Training Loss	0.89
Epochs	3

7. Challenges & Learnings

Several practical issues were encountered, including Kaggle GPU setup, dataset loading problems, and Hugging Face authentication errors during model deployment. These issues were resolved by configuring Kaggle settings correctly, updating API tokens, and managing repository permissions properly. The assignment provided valuable hands-on experience with modern MLOps workflows.

8. Project Links

Kaggle Notebook: <https://www.kaggle.com/code/rathodishag25ait2084/notebook0315781c1b>

Hugging Face Model: <https://huggingface.co/IshaIT/distilbert-news-classifier>

GitHub Repository: <https://github.com/isha-rth/ml-ops-assignment2>

W&B; Dashboard: <https://wandb.ai/g25ait2084-iit/mlops-assignment2/>

9. Conclusion

This assignment successfully demonstrated a complete MLOps pipeline using Kaggle, Hugging Face, and Weights & Biases. The project covered model training, tracking, evaluation, deployment, and reproducibility practices widely used in real-world machine learning systems.